# AC444
# VALUATION
# AND SECURITES ANALYSIS

## Week 5 (Session 2)
## INTRODUCTION TO VALUATION

THE LONDON SCHOOL OF ECONOMICS AND POLITICAL SCIENCE

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# TOPICS DISCUSSED

- Motivations for Valuation;
- Valuation Methods Used by Analysts;
- Valuation Module Course Outline;
- The Method of Comparables;
- Mini Case-Study: AB InBev;
- View from a PE Perspective;
- Economics of Value Creation.

# MOTIVATIONS FOR VALUATION

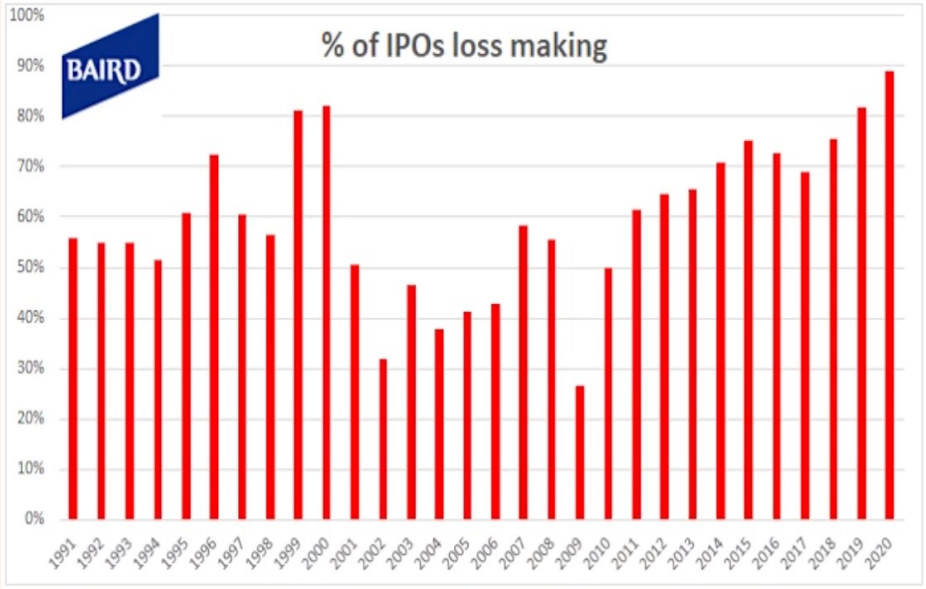
# MOTIVATIONS FOR VALUATION
## Private Firms

- Understanding how to value a firm is crucial when considering providing seed capital or venture capital to start ups;

- It is also crucial in the context of Initial Public Offerings (IPOs).

# MOTIVATION FOR VALUATION
## IPOs

- High stakes;
- But … many IPOs are loss making;
- Important to distinguish between "unviable" and "temporarily unprofitable" IPO firms:
  - ✓ need for analysis;
  - ✓ valuation challenges.

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US Listings

# MOTIVATION FOR VALUATION
## IPOs

- Consider internet consulting firms which went through an IPO between 1999 and 2000:

| Company | IPO Price ($) | Peak Price ($) | % Change IPO to Peak | Peak Date | Feb 28th 2001 Price ($) | % Change From Peak |
|---|---|---|---|---|---|---|
| Scient | 10 | 133.75 | 1,238 | March 2000 | 2.94 | -97.8 |
| Viant | 8 | 63.56 | 695 | December 1999 | 3.06 | -95.2 |
| IXL | 12 | 58.75 | 390 | January 2000 | 1.25 | -97.9 |
| Lante | 20 | 87.50 | 338 | February 2000 | 1.81 | -97.9 |
| Razorfish | 8 | 56.94 | 612 | February 2000 | 1.16 | -98.0 |
| US Interactive | 10 | 83.75 | 738 | January 2000 | 0.56 | -99.3 |

- These firms went trough amazing peak market values yet ultimately ended up being close to worthless…

# MOTIVATION FOR VALUATION IPOs

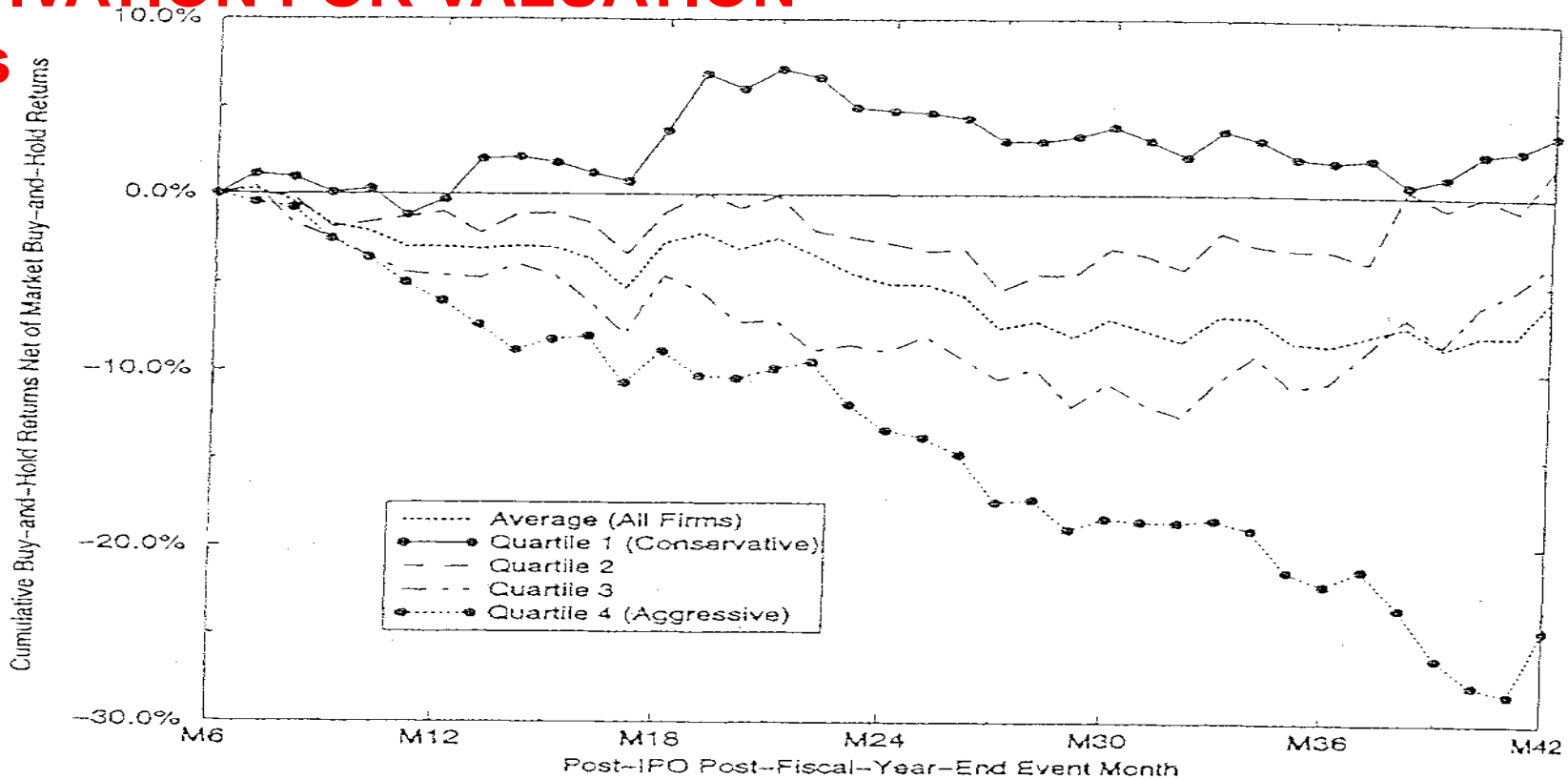


**Figure 2. Cumulative buy-and-hold return net of Nasdaq composite index return of DCA quartiles by event month, in percent.** The sample is of 1,649 IPOs from the 1980–92 period. (Details are described in Table I.) Firms are divided into quartiles based on how aggressively they manage their earnings. Our measure of earnings management is the firm's discretionary current accruals (DCA), which adjusts for industry, size, and growth. Details are described in Table II and Section I.B.

# MOTIVATION FOR VALUATION
## IPOs

- Teoh, Wong, and Rao (Are Accruals during IPOs Opportunistic?) argue that there are:

  - ✓ High uncertainties about newly public firms;
  - ✓ High information asymmetries of information between issuers and investors;
  - ✓ Few reliable independent sources of information about the newly public firms;

  High incentives for firms to manage earnings opportunistically by reporting high IPO accruals!

# MOTIVATION FOR VALUATION
## IPOs

- Teoh, Wong, and Rao (Are Accruals during IPOs Opportunistic?) provide the following empirical evidence:
  - Reported earnings performance of IPO firms is unusually high in the IPO year (compared with that of non-issuing industry peers);
  - Post IPO, IPO firms earn significantly less than non-issuing industry peers;

⇒IPO firms report high earnings during the IPO by reporting abnormal accruals aggressively!

⇒ High abnormal accruals however can not be sustained!

# MOTIVATION FOR VALUATION
## IPOs

- Teoh, Wong, and Rao (Are Accruals during IPOs Opportunistic?) thus report that abnormal accruals, specifically abnormal current accruals, explain the post-issue underperformance in earnings!

- Is it possible to trade successfully on the basis of the magnitude of abnormal accruals in IPO years?

# MOTIVATION FOR VALUATION IPOs

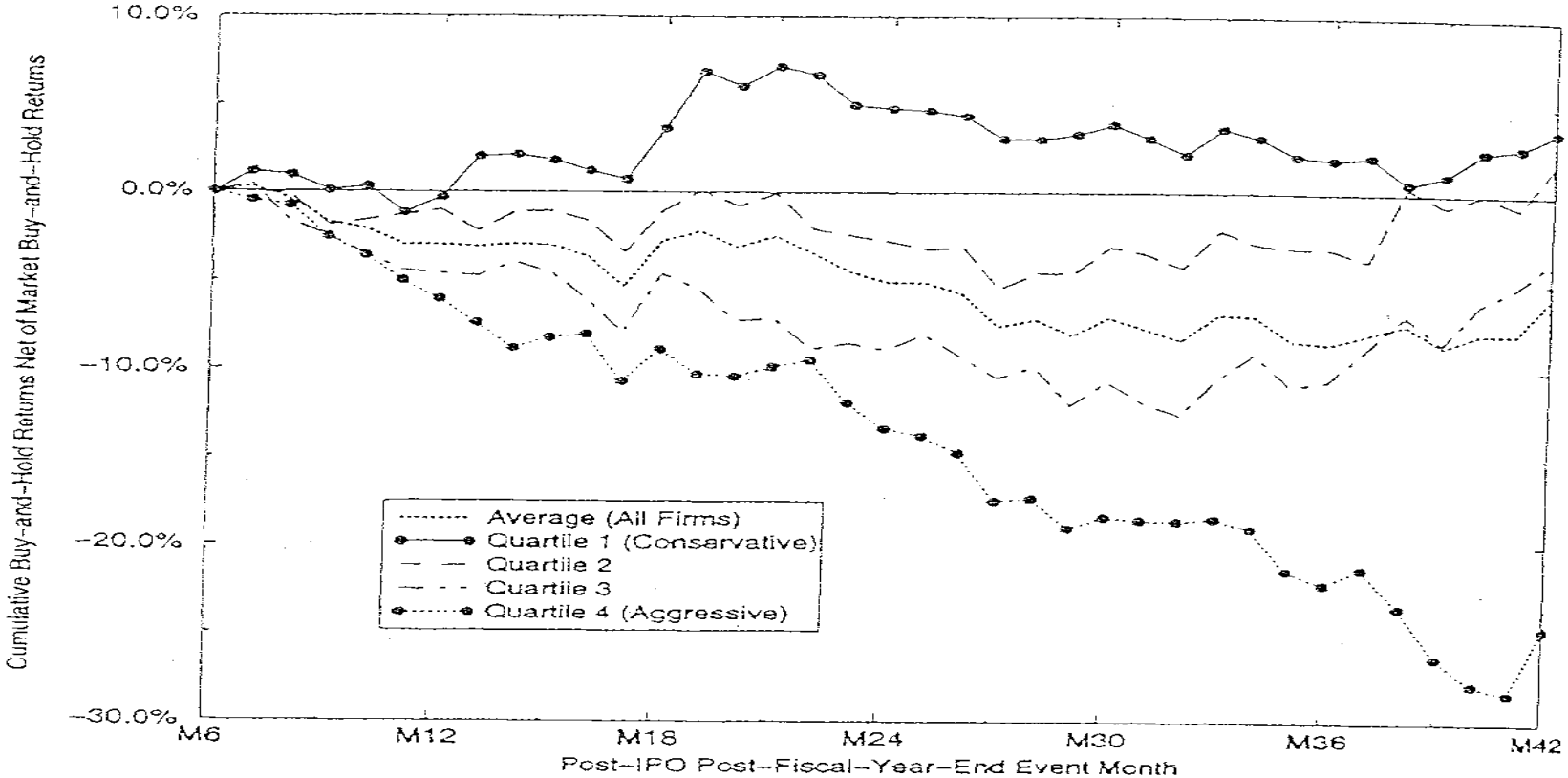


**Figure 2. Cumulative buy-and-hold return net of Nasdaq composite index return of DCA quartiles by event month, in percent.** The sample is of 1,649 IPOs from the 1980–92 period. (Details are described in Table I.) Firms are divided into quartiles based on how aggressively they manage their earnings. Our measure of earnings management is the firm's discretionary current accruals (DCA), which adjusts for industry, size, and growth. Details are described in Table II and Section I.B.

# MOTIVATION FOR VALUATION
## IPOs

Empirical Evidence

- Issuers with unusually high accruals in the IPO year experience poor stock return performance in the 3 years thereafter;
  - →IPO issuers in the most aggressive quartile of earnings managers experience on average a 15 to 30% worse subsequent 3-year performance than IPO issuers in the most conservative quartile;

- Issuers in the most conservative quartile also return to the capital market for a seasoned equity offering about 20% more frequently over a 5 year period than those in the most aggressive quartile;
  - →Post issue benefit to less aggressive IPO issuers.

# MOTIVATION FOR VALUATION
## Public Firms

- Understanding how to value a firm is also important when considering investing in listed companies;
- One cannot assume that capital markets are always efficient, that is, that stocks' market values are equal to their fundamental values;
- To make investment decisions, one may hence wish to:
  - ✓ compute fundamental (also referred to as intrinsic or true) values and compare them with market values to make investment decisions;
  - ✓ engage in reverse engineering.

# MOTIVATION FOR VALUATION
## Public Firms

| Stocks | High (2000) | Low (2002) | % Decline |
|---|---|---|---|
| Amazon.com | 75.25 | 5.51 | 98.7 |
| Cisco Systems | 82.00 | 11.04 | 86.5 |
| Corning | 113.33 | 2.80 | 99.0 |
| JDS Uniphase | 297.34 | 2.24 | 99.5 |
| Lucent Technologies | 74.93 | 1.36 | 98.3 |
| Nortel Networks | 143.62 | 0.76 | 99.7 |
| Priceline.com | 165.00 | 1.80 | 99.4 |
| Yahoo.com | 238.00 | 8.02 | 96.4 |

# MOTIVATION FOR VALUATION
## Price Multiples

- Valuations (price multiples) vary through time;
- Incentives for:
  - ✓ carrying fundamental analysis;
  - ✓ tactical asset allocation.

# SNOWFLAKE IPO

- Snowflake Inc., a large public cloud company (3rd largest), went through an IPO on September 16th, 2020;

- According to its IPO filing, Snowflake Inc. was a cloud-based warehousing company providing:
  - ✓ businesses with an integrated data management system;
  - ✓ support for data engineering, data warehousing, and data science.

- Prior to the IPO:
  - ✓ revenue grew by 189% to $148.9m in the year ended July 3rd, 2020;
  - ✓ net loss decreased from $177.2m to $171.3m.

# SNOWFLAKE IPO

- Key valuation multiples at IPO:
  - ✓ EV/Revenue: 84.8
  - ✓ EV/EBITDA: 159.7

- The IPO (largest software IPO ever as of 2020):
  - ✓ raised $3.4bn at a $120 price per share (market price cap: $33.3bn);
  - ✓ was subscribed by Berkshire Hathaway ($570m);
  - ✓ was oversubscribed by 100%;

- Dual class shares.

Snowflake Inc. + Add to myFT

## Snowflake doubles in first trades after largest-ever software IPO

Cloud computing company raises $3.4bn and market value jumps to almost $90bn

Snowflake's listing is further evidence of the surging demand for tech companies, including cloud computing service groups, benefiting from the shift to remote working © Getty Images

**Miles Kruppa** in San Francisco and **Richard Henderson** in New York

Published SEP 16 2020 | Updated SEP 16 2020, 18:36 72

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Shares in Snowflake doubled on their Wednesday trading debut after the largest initial public offering ever for a US software company, in the latest sign of Wall Street's huge demand for cloud computing service businesses.

The Silicon Valley company's stock jumped more than 160 per cent above the IPO price to touch $315, sending the company's market value to almost $90bn, based on the number of shares outstanding, before easing to $253.93 at the close in New York.

Snowflake raised a larger than planned $3.4bn by selling 28m shares for $120 apiece, exceeding its targeted range of $100 to $110, which itself was increased by more than a quarter earlier this week.

# SNOWFLAKE IPO

- Snowflake Inc.'s stock price rose to:
  - ✓ $253.9 (+112%) by the end of the first day of trading;
  - ✓ $401.9 (+235%) on November 16th, 2021.

- Its EV/Revenue (NTM) rose to 120 by January 2022.

Opinion **The Top Line**

## Goldman Sachs did not misprice Snowflake's Buffett-backed IPO

It is wrong to assume the cloud computing company was hoodwinked by Wall Street just because of its share price pop

**TOM BRAITHWAITE** + Add to myFT

Snowflake raised $3.4bn in its initial public offering and is now worth more than $60bn © Brendan McDermid/Reuters

**Tom Braithwaite**

Published SEP 18 2020

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At Goldman Sachs, they are back in the office and playing drunken games of pin the tail on the donkey.

At least, that is one explanation for the apparent mispricing of the world's biggest ever software IPO this week. Snowflake, backed by Warren Buffett and whose lead underwriter was Goldman, was priced at $120 a share before surging to close the first day at $253.93, 111.6 per cent higher.

# SNOWFLAKE'S FALL TO EARTH

- From the end of 2021, Snowflake Inc.'s stock price fell to:
  - ✓ $113.3 by June 13th, 2022;
  - ✓ $110.4 by September 3rd, 2024.

- Reasons for the fall:
  - ✓ decreases in experienced and expected growth and margins;
  - ✓ insider sales (insider dealing);

Feb 28, 22:03

Jaren Kerr in New York

## Snowflake shares plunge more than 20% on CEO switch and weak outlook

Snowflake shares plunged by more than a fifth after the cloud computing group announced its chief executive was retiring with immediate effect and released a weak outlook in its latest results.

Sridhar Ramaswamy, the company's senior vice-president of artificial intelligence and a former Google executive, has been appointed as the new CEO, replacing Frank Slootman, who had led the company since 2019. Slootman, 65, continue to chair Snowflake's board.

Alongside the unexpected leadership development on Wednesday, Snowflake projected product revenue between $745mn and $750mn in the current quarter, below Wall Street estimates for $765mn.

Snowflake shares were down 23 per cent in after-hours trading, but at one point had shed about one-quarter of their value to hit a three-month low.

# VALUATION METHODS USED BY ANALYSTS

Source:
Evidence on the Use of Valuation Models by Practitioners
(Pinto et al, 2018)

# VALUATION METHODS SELECTED BY FINANCIAL ANALYSTS AND PORTFOLIO MANAGERS

**TABLE 1** Valuation approaches: global ranking. In this and many following questions, the respondents are asked to select all responses that apply. They are given the opportunity to give written responses whenever they select "other." Because respondents frequently use more than one approach, the percent of respondents using the approaches sums to more than 100%. In the final column, given that a respondent uses an approach, he or she is asked for the percentage of valuation cases in which the approach is used. Thus, this column reports conditional frequencies

| In evaluating individual equity securities, which of the following approaches to valuation do you use? ($N$ = 1,980) | Percent of respondents | Percentage of cases respondents use each approach (mean) |
|---|---|---|
| A market multiples approach | 92.8 | 68.6 |
| A present discounted value approach | 78.8 | 59.5 |
| An asset-based approach | 61.4 | 36.8 |
| A (real) options approach | 5.0 | 20.7 |
| Other approach | 12.7 | 58.1 |

# VALUATION METHODS SELECTED BY FINANCIAL ANALYSTS AND PORTFOLIO MANAGERS

**TABLE 2** Market multiples approach: global evidence. In Table 1, 92.8% of respondents report using market multiples. This table details which market multiples these respondents use, and the conditional percentage of cases where they use each multiple (given that they use the multiple)

| When you use a market multiples approach, which of the following ratios do you use? ($N$ = 1,765) | Percent of respondents | Percentage of cases respondents use each approach (mean) |
|---|---|---|
| D/P (Dividend yield) or P/D (Price-to-dividend) | 35.5 | 44.3 |
| Enterprise value (EV) or firm value multiples (e.g., EV-to-EBITDA, EV-to- operating profit) | 76.7 | 61.1 |
| P/B (Price-to-book value, price-to-adjusted book value, book-to-market) | 59.0 | 44.8 |
| P/CF (Price to some measure of cash flow) | 57.2 | 54.6 |
| P/S (Price-to-sales or revenues) | 40.3 | 45.7 |
| P/E (Price to some measure of earnings) | 88.1 | 67.2 |
| Other ratios | 11.6 | 58.5 |

# VALUATION METHODS SELECTED BY FINANCIAL ANALYSTS AND PORTFOLIO MANAGERS

**TABLE 3** Definition of earnings in P/E multiples. In the denominator of P/E ratios, do analysts prefer trailing income (reported income for the last four quarters) or forecasted income for the next four quarters (forecasted by analysts)?

| When you use a P/E multiple, what measure of earnings do you prefer to use in the denominator? ($N$ = 1,505) | Percent of respondents |
|---|---|
| Trailing net income | 8.8 |
| Trailing operating income | 4.5 |
| Forecasted net income | 61.1 |
| Forecasted operating income | 20.1 |
| Other | 5.5 |

# VALUATION METHODS SELECTED BY FINANCIAL ANALYSTS AND PORTFOLIO MANAGERS

**TABLE 4** Measures used in denominator of enterprise value multiples. For those respondents who use enterprise value multiples, which measure do they prefer in the denominator?

| When you use an enterprise value (EV) multiple, which of the following measure(s) do you use in the denominator? ($N$ = 1,308) | Percent of respondents |
|---|---|
| EBIT | 19.3 |
| EBITDA | 88.3 |
| Free cash flow to the firm | 21.2 |
| Revenue | 16.6 |
| Other | 5.6 |

# VALUATION METHODS SELECTED BY FINANCIAL ANALYSTS AND PORTFOLIO MANAGERS

**TABLE 5** Definition of cash flow in P/CF multiples. For those respondents who use P/CF multiples, which measure of cash flow do they prefer in the denominator?

| When you use a P/CF multiple, which measure of cash flow do you prefer to use in the denominator? ($N$ = 967) | Percent of respondents |
|---|---|
| Operating cash flow | 22.3 |
| Free cash flow to the firm | 28.9 |
| Free cash flow to equity | 32.2 |
| EBITDA | 12.7 |
| Other | 3.9 |

**TABLE 6** Present discounted value approaches: global evidence. In Table 1, 78.8% of respondents report using discounted cash flow models. This table details which DCF models these respondents use, and the conditional percentage of cases where they use each model (given that they use the DCF model)

| When you use a present discounted value model approach, which of the following models do you use? ($N$ = 1,457) | Percent of respondents | Percentage of cases respondents use each approach (mean) |
|---|---|---|
| Dividend discount model | 35.1 | 51.7 |
| Residual income approach (e.g., based on discounted abnormal earnings, economic profit, EVA, or similar concepts) | 20.5 | 46.1 |
| Discounted free cash flow model | 86.8 | 80.1 |
| Cash flow return on investment (CFROI®) model | 19.7 | 58.5 |
| Other | 3.6 | 71.3 |

# COURSE OUTLINE

# COURSE OUTLINE

- Valuation:

  - ✓ Week 5 (Session 2):
    Motivations for Valuation;
    Valuation of Start-Ups and Pre-IPO companies;
    The Method of Comparables;
    Application to the IPO of Snap Inc.

  - ✓ Week 7 (Session 1):
    FCFE-Based and Accounting-Based Valuations.

  - ✓ Week 7 (Session 2):
    Implications for Price Multiples;
    Strategic Taxonomy and Price Multiples.

# COURSE OUTLINE

- Valuation:
  - ✓ Week 8 (Session 1):
    Cash Flow-Based Valuations;
    Case-Study: Shamrock Capital: Pricing the Masters of Taylor Swift.
  - ✓ Week 8 (Session 2):
    Forecasting;
    Structure and Implementation of Valuations;
    Case-Study: L'Oreal: Recommendation on the Share Price.
  - ✓ Week 9 (Session 1):
    Estimating the Cost of Capital;
    Estimating Continuing Values;
    Case-Study: L'Oreal: Recommendation on the Share Price.

# COURSE OUTLINE

- Valuation:
  - ✓ Week 9 (Session 2):
    Case-Study: Snap Inc.'s IPO (A).
  - ✓ Week 10 (Session 1):
    Valuation in the Context of M&As;
    Case-Study: Sanofi-Aventis' Tender Offer for Genzyme.
  - ✓ Week 10 (Session 2):
    Case-Study: Sanofi-Aventis' Tender Offer for Genzyme.
  - ✓ Week 11:
    Project Presentations.

# METHOD OF COMPARABLES

# METHOD OF COMPARABLES

- Assume we wish to value firm A about to go through an IPO;

- The method of comparables involves the following 4 steps:
  1. Selecting a fundamental of A, $F_A$, such as NP, Sales, CF, CSE, or NOA;
  2. Identifying a comparable firm B;
  3. Deriving the price-to-fundamental, $PF_B$, for the comparable firm;
  4. Applying the comparable firm's price-to-fundamental to firm A's selected fundamental to generate an intrinsic value $V_A$:

     $V_A = F_A * PF_B$

# METHOD OF COMPARABLES

- Intuition:
  - ✓ Firms A and B are similar and should hence trade on the same price-to-fundamental ratio;

# METHOD OF COMPARABLES

- Main issues:
  - ✓ selection of fundamental;
  - ✓ selection of comparable firms.

# PERCENTILES OF COMMON PRICE MULTIPLES

| Percentile | PB | PE (Trailing[a]) | PE (Leading[b]) | PS | PCFO |
|---|---|---|---|---|---|
| 95 | 7.4 | Negative Earnings | 41.7 | 4.1 | Negative CFO |
| 75 | 2.5 | 29.4 | 19.2 | 1.7 | 21.9 |
| 50 | 1.5 | 17.5 | 14.3 | 0.8 | 10.0 |
| 25 | 0.9 | 12.3 | 10.9 | 0.4 | 5.8 |
| 5 | 0.5 | 7.6 | 7.3 | 0.2 | 2.5 |

**Percentiles of common price multiples for US-listed firms**

a. A trailing PE ratio is defined as the ratio of the market value of equity over the last reported earnings.
b. A leading PE ratio is defined as the ratio of the market value of equity over one-year-ahead earnings forecasts.

# PRICE MULTIPLES
# IN THE EUROPEAN AIRLINE INDUSTRY (2024)

| Name | Market Cap (US$) | PE | ROE (%) | D/E (%) | PB | PCF | Beta |
|---|---|---|---|---|---|---|---|
| Ryanair | 25,527,645,407 | 11.55 | 28.8% | 72.9% | 2.98 | 6.99 | 1.49 |
| IAG | 9,789,609,322 | 3.59 | 100.4% | 490.6% | 2.71 | 2.00 | 2.89 |
| Lufthansa | 8,896,504,957 | 4.25 | 21.3% | 143.7% | 0.85 | 2.31 | 1.34 |
| Easyjet | 5,151,608,738 | 13.04 | 12.2% | 103.5% | 1.52 | 3.25 | 3.17 |
| Jet2 | 3,828,759,518 | 7.97 | 29.8% | 135.8% | 1.96 | 3.87 | 2.93 |
| AF- KLM | 2,748,623,236 | 24.50 | | 2,705.0% | -0.81 | 1.01 | 1.96 |
| Wizzair | 2,729,507,789 | 14.30 | 1,953.6% | | 10.01 | 2.47 | 2.63 |
| Norwegian | 1,511,781,289 | 10.81 | 34.8% | 225.3% | 2.78 | 2.04 | 1.16 |
| Finnair | 677,851,168 | 1.35 | 54.9% | 351.0% | 1.08 | 0.60 | 1.37 |

# METHOD OF COMPARABLES

- Trading vs Transaction multiples;
  - ✓ Transaction multiples include control/takeover premia.

- Enterprise value-based vs Equity-based multiples;
  - ✓ Enterprise value-based multiples are less influenced by capital structure and earnings such as EBITDA or EBIT less likely to be negative than Net Income.

- Historical vs Forward-Looking multiples;
  - ✓ Empirical evidence suggests that the latter ones tend to be more accurate (eliminates issues with one-off events).

# METHOD OF COMPARABLES

- Using a sample of comparable firms can be better than a single comparable.

- EBITDA multiples tend to be better predictors of value compared with. EBIT or Revenue multiples.

# METHOD OF COMPARABLES APPLICATION TO AB InBev

# METHOD OF COMPARABLES APPLICATION TO AB InBev

| AB InBev | |
|---|---|
| | |
| (€m) | **2018** |
| Revenue | 54619 |
| Cost of Goods Sold | 17451 |
| Gross Profit | 37168 |
| Operating Expenses | 16264 |
| EBITDA | 20904 |
| Depreciation & Amortization | 2908 |
| EBIT | 17996 |
| Financial Expense | 9401 |
| EBT | 8595 |
| Tax Expense | 2922 |
| Net Income | 5673 |
| | |
| Book Value of Equity (CSE) | 71904 |
| Net Debt (NFO) | 118700 |
| Net Operating Assets (NOA) | 190604 |
| | |
| # Shares | 2019 |

| | PE | EV/EBITDA | EV/EBIT | EV/Revenue | P/CSE |
|---|---|---|---|---|---|
| AB inBev | 19.3 | 12.7 | 15.9 | 5.4 | 2.6 |
| Heineken | 20.5 | 12.3 | 17.2 | 2.9 | 3.6 |
| Carlsberg | 22.8 | 11.5 | 16.0 | 2.6 | 3.4 |
| Diageo | 24.0 | 19.1 | 21.1 | 6.8 | 9.9 |

# METHOD OF COMPARABLES APPLICATION TO AB InBev

- The Task:

  What is your best estimate of AB InBev' s intrinsic share price?

# MULTIPLES VALUATION AND ITS USE IN THE PRIVATE EQUITY INDUSTRY

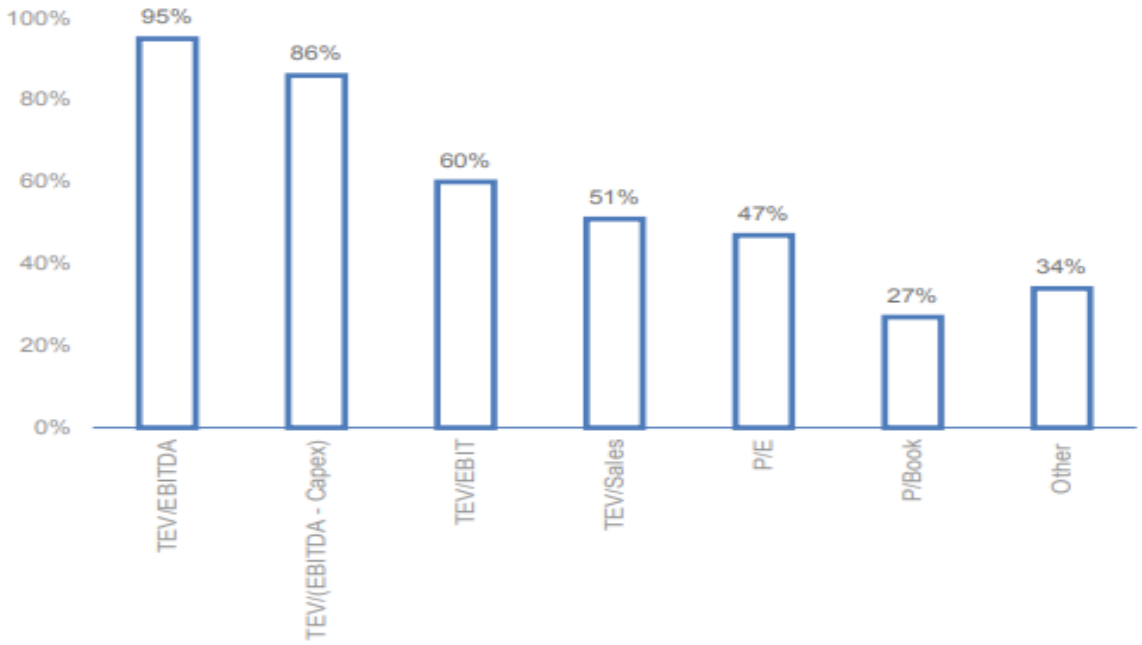
Source:

Primer on Multiple Valuation and its Use in the Private Equity Industry

[Ivashina and Boe (HBS)]

# Most Used Multiples By Private Equity Firms

**Figure 1** Valuation Multiples Used by Private Equity Firms and Their Typical Use




Source: Compiled by casewriters based on discussions with current and former employees of 10 large private equity firms.

Note: Numbers indicate prevalence of use based on the surveyed sample. For example, 95% of surveyed firms use TEV/EBITDA multiples. Use of different multiple approaches is not mutually exclusive: the figure indicates that more than half of the surveyed firms use both EBITDA and EBIT multiples.

# METHOD OF COMPARABLES

## How Are Multiples Linked to DCF Models?

- Consider a firm and assume that its free cash-flows (FCF) are expected to grow at g per year;

- This company's enterprise value should thus be valued at:

$$EV_t = \frac{FCF_{t+1}}{r - g}$$

with:

- $EV_t$ denoting the firm's Enterprise Value (intrinsic value of the firm's Net Operating Assets (NOA);
- r denoting the cost of the firm's capital.

# METHOD OF COMPARABLES

## How Are Multiples Linked to DCF Models?

- It follows that NTM (Next Twelve Months) and LTM (Last Twelve Months)-based multiples can be derived as:

$$\frac{EV_t}{FCF_{t+1}} = \frac{1}{r-g}$$

$$\frac{EV_t}{FCF_t} = \frac{1+g}{r-g}$$

- Issues:
  - ✓ Growth rates fluctuate due to cyclicality and other risk factors;
  - ✓ FCFs thus can be very volatile and lumpy:
    - ✓ Large one-of capital expenditure;
    - ✓ Acquisition;
    - ✓ Large working capital fluctuations.
  - ✓ Many firms also have negative FCFs.

# METHOD OF COMPARABLES

## How Are Multiples Linked to DCF Models?

- Hence, one could:
  - ✓ Make adjustments to the FCF to better capture the true earnings potential of the company;
  - ✓ Move up in the company's operations to find a metric allowing for better comparability.

# METHOD OF COMPARABLES

## How Are Multiples Linked to DCF Models?

### EBITDA

FCF = (1-T) EBITDA + T*DEP – ΔNWC – CAPEX

$$\frac{\text{EV}}{\text{EBITDA}}$$

EBITDA:

- ✓ is less lumpy than FCF;
- ✓ is less likely to be negative than FCF;
- ✓ allows for comparability when different firms use different depreciation methods;
- ✓ does not account for CAPEX but can be adjusted for this (e.g. EBITDA – CAPEX);
- ✓ can be further adjusted for differences in which firms account for rent through EBITDAR.

### EBIT

FCF = (1-T) EBIT + DEP – ΔNWC – CAPEX

$$\frac{\text{EV}}{\text{EBIT}}$$

EBIT:

- ✓ is less lumpy than FCF;
- ✓ is less likely to be negative than FCF;
- ✓ accounts for differences in capital intensity through depreciation;
- ✓ does not account for differences in treatment of depreciation or amortization across firms.

# OPERATIONAL AND FINANCIAL METRICS FREQUENTLY USED BY THE PE INDUSTRY

| Metric | When and How to Use |
|---|---|
| **Operational Metrics** | Nascent firms with limited financial data (e.g. VC backed companies).<br>Far from FCF (potentially large variations between operational metrics and FCF). Will generate rough approximations of value. |
| **Sales** | Early stage and growth stage firms.<br>Useful when earnings are negative due to cyclical downturn.<br>Far from FCF (potentially large variations between sales and FCF). Will generate rough approximations of value. |
| **EBITDA** | Stable PE metric suitable for mature firms.<br>Can include industry specific adjustments (e.g. EBITDAR).<br>Fairly robust. |
| **EBITDA – Maintenance CAPEX** | Helpful when capital intensity may vary between firms.<br>Must be estimated though. |
| **EBITDA - CAPEX** | Common metric, second only to EBITDA.<br>Adjusts for capital intensity.<br>One could try to normalize CAPEX to avoid lumpiness. |
| **EBIT** | Standard financial metric, easy to find.<br>Susceptible to differing accounting treatments of depreciation and amortization. |
| **FCF** | Ideal metric.<br>Seldom available or at steady state though. |

# VALUE CREATION

# DRIVERS OF FCF-BASED VALUATIONS

- We know that:

$$FCF = NOPAT - \Delta NOA$$

where:

  - ✓ NOA captures the book value of Net Operating Assets;
  - ✓ ΔNOA capture the net new investment (CAPEX).

- Define the investment ratio as:

$$IR = \frac{\Delta NOA}{NOPAT}$$

# DRIVERS OF FCF-BASED VALUATIONS

- Define the firm's Return on Net Operating Assets (RNOA) as:

$$RNOA = \frac{NOPAT}{NOA}$$

- Assuming that FCF and NOPAT are expected to grow by g, then:

$$g = RNOA * IR$$

- It follows that:

$$FCF = NOPAT\left(1 - \frac{g}{RNOA}\right)$$

# DRIVERS OF FCF-BASED VALUATIONS

- The firm's Enterprise Value (intrinsic value of NOA) is thus:

$$EV_t = \frac{FCF_{t+1}}{r - g} = \frac{NOPAT_{t+1}\left(1 - \frac{g}{RNOA}\right)}{r - g}$$

- The latter valuation is referred to as the FCF-based valuation driver formula;

- Future profitability and growth hence drive DCF valuations.

# DRIVERS OF FCF-BASED VALUATIONS
## Example

- Consider 2 companies differing only in terms of expected RNOA (and hence IR):

| Company | $NOPAT_{t+1}$ ($m) | g | RNOA | r | $EV_t$ ($m) |
|---|---|---|---|---|---|
| Value Inc. | 100 | 5% | 20% | 10% | ? |
| Volume Inc. | 100 | 5% | 10% | 10% | ? |

- Which company has the highest enterprise value?

# DRIVERS OF FCF-BASED VALUATIONS
## Example

| Company | $NOPAT_{t+1}$ ($m) | g | RNOA | r | $EV_t$ ($m) |
|---|---|---|---|---|---|
| Value Inc. | 100 | 5% | 20% | 10% | 1,500 |
| Volume Inc. | 100 | 5% | 10% | 10% | 1,000 |

$$EV_t = \frac{\$100m\left(1-\frac{5\%}{20\%}\right)}{10\%-5\%} = \$1{,}500\text{m}$$

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# DRIVERS OF FCF-BASED VALUATIONS

- Given that:

$$EV_t = \frac{NOPAT_{t+1}\left(1 - \frac{g}{RNOA}\right)}{r - g}$$

- A firm's leading Enterprise Value to NOPAT can be written as follows:

$$\frac{EV_t}{NOPAT_{t+1}} = \frac{1 - \frac{g}{RNOA}}{r - g}$$

- A firm's Enterprise Value to NOA can be written as follows:

$$\frac{EV_t}{NOA_t} = RNOA\left[\frac{1 - \frac{g}{RNOA}}{r - g}\right]$$

- Future profitability and growth thus drive a firm's (leading) price-to-earnings and price-to-book ratios.

# DRIVERS OF FCF-BASED VALUATIONS
## Summary

- There is no reason for g and RNOA to reman constant through time forever;
- This assumption was made to highlight the drivers of DCF-based valuations and enterprise value-based price multiples;
- It is used in practice though when deriving continuing values in DCF-based valuations.

# DRIVERS OF FCF-BASED VALUATIONS
## Empirical Evidence on Profitability

- RNOA is driven by competitive advantage enabling companies to realize:
  - ✓ price premia;
  - ✓ cost efficiencies;
  - ✓ capital efficiencies.

- Industry structure is an important determinant of RNOA:
  - ✓ some industries are more likely to generate high returns
  - ✓ there are still significant intra-industry variations though.

- Firms currently enjoying high RNOA are likely to subsequently enjoy high RNOA:
  - ✓ over time;
  - ✓ through changing conditions especially in industries enjoying relatively long product life cycles.

- There is evidence of mean-reversion though.

# DRIVERS OF FCF-BASED VALUATIONS
## Empirical Evidence on Growth

- For large companies:
  - ✓ growth in the markets in which they operate largely drives long-term revenue growth;
  - ✓ gains in market share are less important for long-term growth.

- Wrt valuation, the value created by an incremental $ of revenue is crucial;

- Growth strategies with highest to lowest potential:
  - ✓ true product innovation;
  - ✓ attracting new customers to an existing product;
  - ✓ persuading existing customers to buy more;
  - ✓ growth through acquisitions;
  - ✓ revenue growth from market share gains.